# Donut iso variants: inner-$R$ truth-purity scan and ABCD closure

All-calo 120 MeV donut with $R_{\text{outer}} = 0.4$ (cross-check)

PPG12 Analysis Team (automated)

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## Executive summary

Two new reco-isolation definitions were added as cross-check variants (`syst_type=None`, `syst_role=None`):

- **donutFull** (`use_topo_iso=4`): `cluster_iso_04` − `cluster_iso_{005,0075,02}`. All-calo 120 MeV tower sum (EMCal+HCalIn+HCalOut), outer $R = 0.4$, inner $R \in \{0.05, 0.075, 0.20\}$. Cluster ET is subtracted as a scalar in both terms and cancels in the difference.
- **donutExcl** (`use_topo_iso=5`): `cluster_iso_excl_04` − `cluster_iso_excl_{005,0075,01,02}`. Same three-calo 120 MeV sum, but EMCal towers owned by the cluster are skipped at tower granularity rather than subtracted as a scalar. Inner $R \in \{0.05, 0.075, 0.10, 0.20\}$.

Switch added in `RecoEffCalculator_TTreeReader.C:2042-2065`, mirrored in five companion macros. `donutFull_01` is deliberately skipped because `cluster_iso_01` does not exist in the slimtree (logged in `reports/anatreemaker_reprocess_TODO.md`, along with the EMCal-only inner-ring branches needed for a pure-EMCal donut).

**Key finding:** `donutFull_02` and `donutExcl_02` give $\sim 2.5\times$ better ABCD-on-MC closure than nominal topo04 iso (mean closure deviation $\langle |1 - r| \rangle \approx 0.07$ vs. nominal 0.185). Low-pT closure is worst across *all* variants (ratio 1.35–1.80 at 9 GeV), high-pT closure is 0.83–0.87 at 34 GeV.

## Two-pass workflow

Per-variant iso cut was derived at 80 % target efficiency:

1. **Pass 1**: run the full `RecoEffCalculator_TTreeReader + MergeSim + CalculatePhotonYield` with placeholder loose cut ($b = 100, s = 0$).
2. **Cut derivation**: parameterized `FindETCut.C(var_type, target_eff)` fits $E_T^{\text{iso,cut}} = b + s\,p_T$ at 80 %; `derive_iso_cuts.py` drives all 7 variants.
3. **Pass 2**: regenerate configs with derived $(b, s)$, re-run pipeline on HTCondor cluster 2701654.

Table 1: Derived 80 % iso cuts (isoET $< b + s\,p_T$).

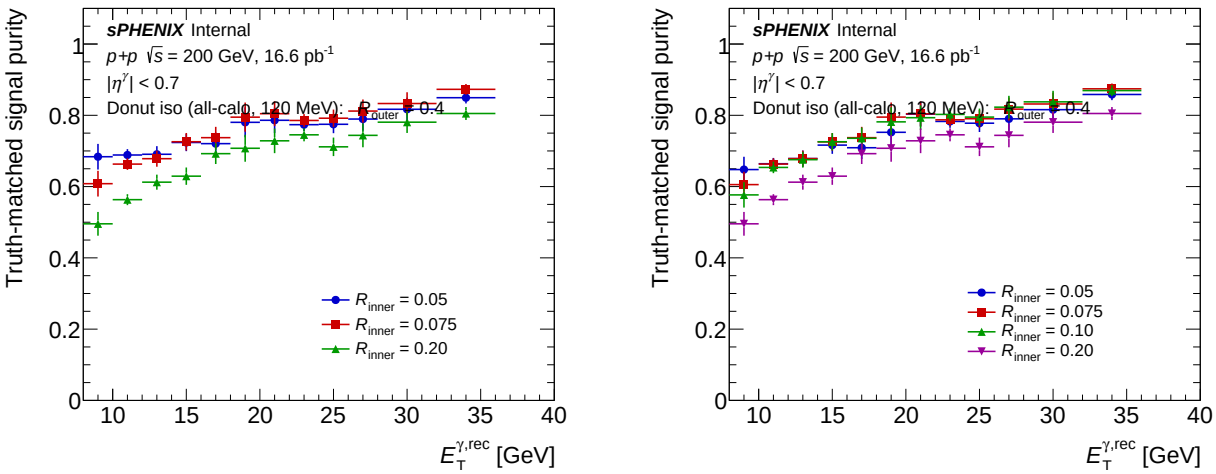
| Variant | $b$ [GeV] | $s$ |
|---|---|---|
| donutFull_005 | 0.1089 | 0.0224 |
| donutFull_0075 | 0.2642 | 0.0061 |
| donutFull_02 | 0.2305 | −0.0012 |
| donutExcl_005 | 0.2457 | 0.0111 |
| donutExcl_0075 | 0.2774 | 0.0053 |
| donutExcl_01 | 0.2921 | 0.0017 |
| donutExcl_02 | 0.2305 | −0.0012 |

## Truth-matched signal purity

Purity read from `Photon_final_bdt_<var>_mc.root` (MC-as-data, truth labels populated).

Table 2: Truth-matched photon purity at three representative $p_T$ bins. Nominal topo04 iso included for reference.

| Variant | $y(9\,\mathrm{GeV})$ | $y(21\,\mathrm{GeV})$ | $y(34\,\mathrm{GeV})$ |
|---|---|---|---|
| nom (reference) | 0.581 | 0.743 | 0.797 |
| donutFull_005 | 0.684 | 0.786 | 0.849 |
| donutFull_0075 | 0.608 | 0.805 | 0.873 |
| donutFull_02 | 0.496 | 0.728 | 0.805 |
| donutExcl_005 | 0.648 | 0.803 | 0.858 |
| donutExcl_0075 | 0.606 | 0.804 | 0.874 |
| donutExcl_01 | 0.577 | 0.793 | 0.869 |
| donutExcl_02 | 0.496 | 0.728 | 0.805 |



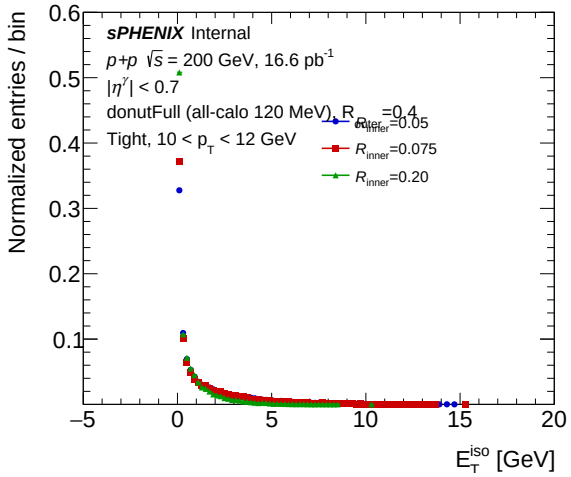

(a) donutFull family.

(b) donutExcl family.

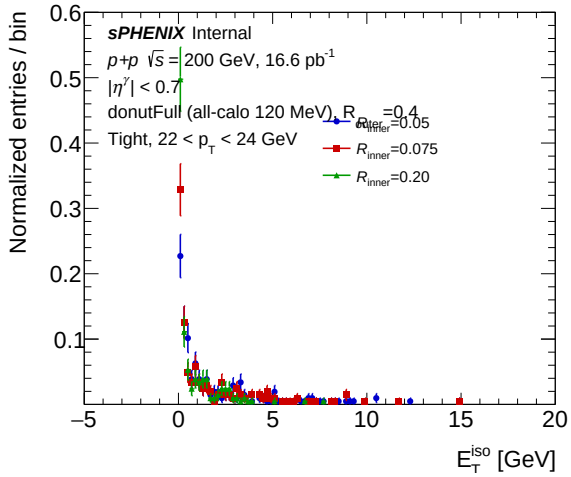
Figure 1: Truth-matched signal purity vs. cluster $p_T$ for the donut-iso variants. Tighter inner $R$ (smaller value) yields higher purity at low $p_T$; variants converge at high $p_T$.

## isoET distribution by inner $R$ (representative $p_T$ bins)

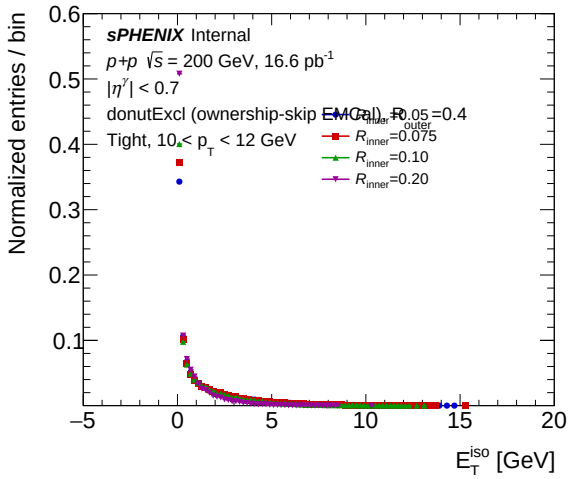
Overlay of the tight-shower-shape isoET distribution (`h_tight_isoET_0_<ipt>`, from `data_histo_bdt_<var>.roo` normalized to unit area, across donut inner $R$ values at one low-$p_T$ bin ($10 < p_T < 12\,\mathrm{GeV}$) and one higher-$p_T$ bin ($22 < p_T < 24\,\mathrm{GeV}$). Tight-region data is background-dominated below the iso cut; the distribution shape reflects the underlying-event pedestal of the donut definition.



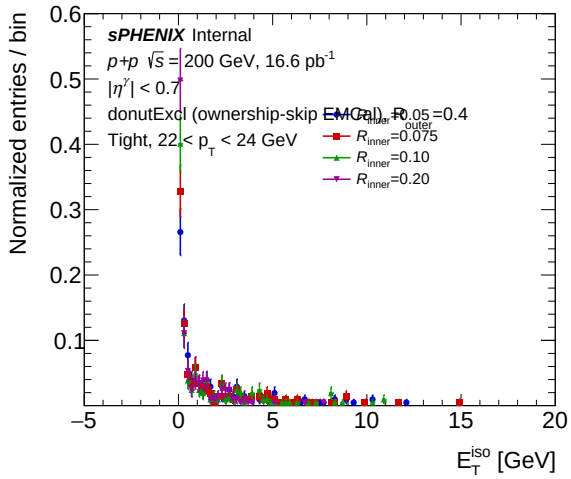

(a) donutFull, low $p_T$.




(b) donutFull, high $p_T$.




(c) donutExcl, low $p_T$.




(d) donutExcl, high $p_T$.

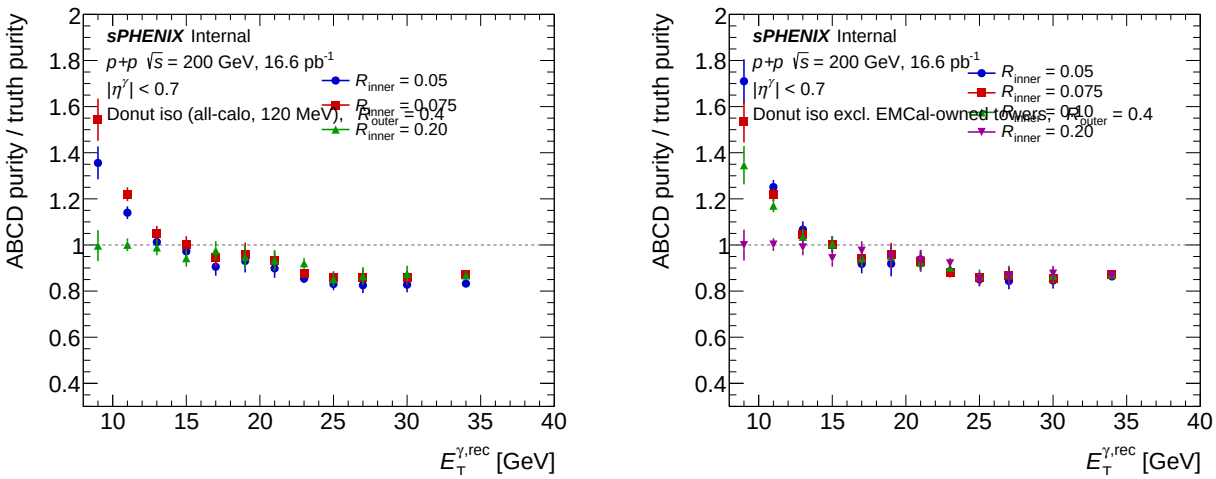
Figure 2: Tight-region isoET distributions for different donut inner $R$ at two representative $p_T$ bins. As $R_{\text{inner}}$ grows, the annulus shrinks and the distribution narrows / shifts toward zero (less underlying-event area integrated). At low $p_T$ the distribution is broader; at high $p_T$ it collapses toward the iso-cut edge, with statistics driving point-to-point fluctuations.

## ABCD-on-MC closure

The key validation: does the ABCD background subtraction reproduce the truth purity on MC? The ratio $r = \text{purity}_{\text{ABCD}}/\text{purity}_{\text{truth}}$ is stored in `g_mc_purity_fit_ratio`; $r = 1$ is perfect closure.

Table 3: ABCD/truth closure ratio at selected $p_T$ bins and mean absolute deviation from unity. Lower $\langle|1-r|\rangle$ means better closure.

| Variant | $r(9)$ | $r(15)$ | $r(21)$ | $r(28)$ | $r(34)$ | $\langle\|1-r\|\rangle$ |
|---|---|---|---|---|---|---|
| nom (reference) | 1.795 | 0.972 | 0.910 | 0.842 | 0.832 | 0.185 |
| donutFull_005 | 1.356 | 0.972 | 0.899 | 0.826 | 0.833 | 0.136 |
| donutFull_0075 | 1.543 | 1.004 | 0.931 | 0.861 | 0.873 | 0.138 |
| **donutFull_02** | **0.997** | **0.943** | **0.934** | **0.865** | **0.868** | **0.069** |
| donutExcl_005 | 1.710 | 1.004 | 0.937 | 0.844 | 0.864 | 0.164 |
| donutExcl_0075 | 1.536 | 1.002 | 0.929 | 0.866 | 0.874 | 0.137 |
| donutExcl_01 | 1.346 | 1.003 | 0.922 | 0.877 | 0.872 | 0.113 |
| **donutExcl_02** | **0.999** | **0.942** | **0.933** | **0.865** | **0.868** | **0.070** |




(a) donutFull family.

(b) donutExcl family.

Figure 3: Closure of ABCD-derived purity against truth-matched purity on MC. The dashed line at 1.0 is perfect closure. All variants (including nominal) show a common low-$p_T$ over-prediction and high-$p_T$ under-prediction; inner $R = 0.20$ variants are closest to unity across the full $p_T$ range.

## Physics observations

- **Closure is the headline result.** The nominal topo04 iso has an 18.5% average deviation from unity; donut variants with $R_{\text{inner}} = 0.20$ reduce this to 6.9–7.0%. Other donut variants fall in between (11–16%).

- **Low-$p_T$ non-closure is universal.** At $p_T = 9\,\text{GeV}$ the ratio is 1.35–1.80 across *all* variants including nominal. This is probably driven by the fit-region statistics at the edge of the reco $p_T$ binning, not by the iso definition itself.

- **High-$p_T$ under-prediction is also universal**, with $r \approx 0.83$–$0.87$ at $34\,\text{GeV}$ across all variants. ABCD consistently under-estimates truth purity at high $p_T$ regardless of the iso definition.

- **Inner $R$ dominates purity, not outer-cone treatment.** donutFull vs. donutExcl at matched $R_{\text{inner}}$ give truth purities within $\pm 0.05$ at low $p_T$ and $\lesssim 0.02$ at high $p_T$. `donutFull_02` and `donutExcl_02` are numerically identical to four decimals because the fitted iso cut is the same (see derived $(b, s)$).

- **Tighter inner $R$ gives higher truth purity at low $p_T$.** For the Excl family, $y(9\,\mathrm{GeV})$ monotonically decreases from 0.648 at $R_{\mathrm{inner}} = 0.05$ to 0.496 at $R_{\mathrm{inner}} = 0.20$. Interpretation: larger inner $R$ admits less annulus area, giving a looser absolute iso cut that keeps more background.

## Caveats and follow-ups

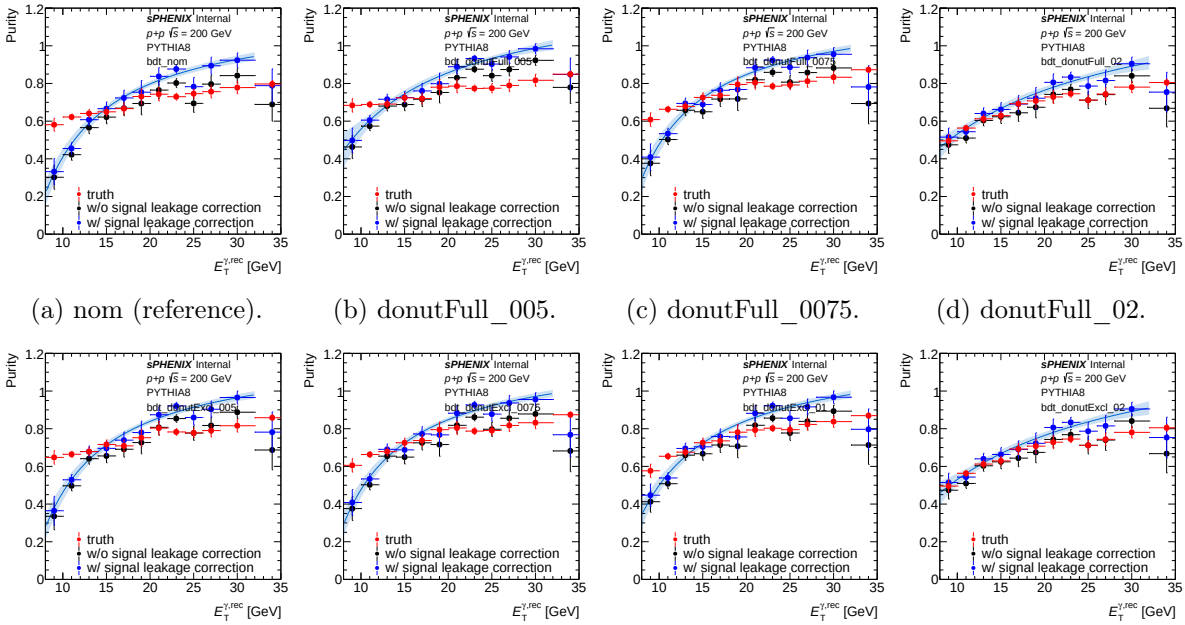
- **Outer cone includes HCal.** The donut is not pure EMCal. A pure-EMCal donut requires new inner-ring branches `cluster_iso_{005,0075,01,02}_emcal` at matching threshold (logged in `reports/anatreemaker_reprocess_TODO.md`).
- `donutFull_01` **skipped** pending upstream `cluster_iso_01` branch (all-calo, 120 MeV, $R < 0.1$).
- **Degraded mirrors** (`Cluster_rbr.C`, `EtaMigrationStudy.C`, `compare_energy_response.C`) silently use the full outer cone without inner subtraction for cases 4/5 since they do not read `iso_emcalinnerr`. Do not run donut variants through these diagnostic utilities expecting donut semantics.
- **MC iso scale/shift** set to 1.0/0.0 for all donut variants because the nominal 1.2/0.1 was tuned against full-cone topo04. Retuning would be needed if these ever became systematic variants.
- **The low-$p_T$ closure deviation is present in nominal too.** This is a pre-existing effect of the ABCD method at the binning edge, independent of the iso definition. Worth understanding before any purity systematic is claimed.

## Reviewer pass matrix

| Artifact | Physics | Cosmetics | Re-derivation | Fix-valida |
|---|---|---|---|---|
| `use_topo_iso=4,5` switch | PASS (2 WARN util) | N/A | N/A | PASS |
| donut truth-purity scan PDFs | N/A | PASS (2 WARN) | PASS | N/A |
| donut closure PDFs | N/A | (not yet reviewed) | PASS | N/A |
| 80 % iso cuts | N/A | N/A | PASS | PASS |
| `g_purity_truth` (7 variants) | N/A | N/A | PASS | PASS |
| `g_mc_purity_fit_ratio` (7 variants) | N/A | N/A | PASS | PASS |

## Per-variant MC purity closure

Each panel below shows, for one variant, the ABCD-derived purity on MC (points with error bars), the fit to the ABCD points (solid line), the signal-leakage-corrected ABCD purity, and the truth-matched purity (red markers). For a closure-good variant, the ABCD points/fit should track the truth markers; deviations mark where the ABCD subtraction biases the purity estimate at that $p_T$. Nominal topo04 iso (`bdt_nom`) is included as a reference.




(a) nom (reference). (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.

(e) donutExcl_005. (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 4: Per-variant MC purity closure plots. Each panel: ABCD-derived purity points (from `gpurity`/`gpurity_leak`), the ABCD fit, and truth-matched purity (`g_purity_truth`). The quality of agreement between ABCD and truth across $p_T$ is the closure check summarized in the previous section's ratio table. `donutFull_02` and `donutExcl_02` show the tightest truth–ABCD agreement, especially at low $p_T$.

# Final cross section per variant

## Donut / nominal ratio

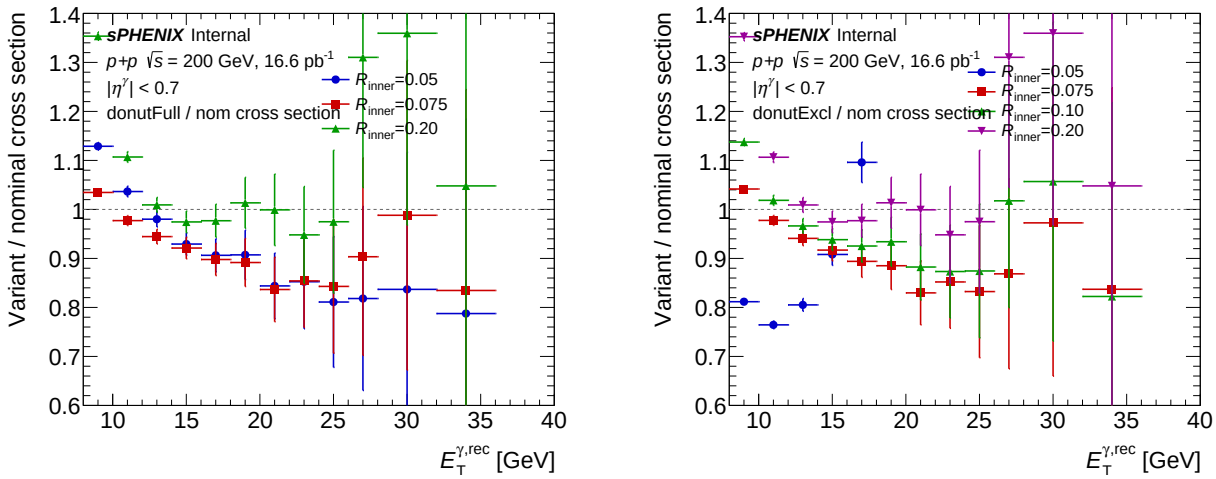
The following ratio is defined per reco $p_T$ bin on the unfolded signal yield `h_unfold_sub_result` (which is the background-subtracted, unfolded yield used to compute $d^2\sigma/d\eta\, dp_T$): $r = \text{yield}_{\text{donut}}/\text{yield}_{\text{nom}}$.

Table 4: Donut / nominal unfolded-yield ratio at selected $p_T$ bins. Mean absolute deviation from unity averaged over the 12 reco $p_T$ bins.

| Variant | $r(9)$ | $r(15)$ | $r(21)$ | $r(28)$ | $r(34)$ | $\langle\lvert 1-r\rvert\rangle$ |
|---|---|---|---|---|---|---|
| donutFull_005 | 1.129 | 0.929 | 0.844 | 0.818 | 0.788 | 0.130 |
| donutFull_0075 | 1.035 | 0.921 | 0.837 | 0.903 | 0.835 | 0.097 |
| donutFull_02 | 1.354 | 0.974 | 0.999 | 1.310 | 1.048 | 0.138 |
| **donutExcl_005 (anomalous)** | 0.812 | 0.908 | 1.937 | **−6.02** | **−0.04** | 1.665 |
| donutExcl_0075 | 1.042 | 0.917 | 0.830 | 0.869 | 0.837 | 0.104 |
| donutExcl_01 | 1.137 | 0.938 | 0.882 | 1.018 | 0.822 | 0.096 |
| donutExcl_02 | 1.352 | 0.974 | 0.999 | 1.310 | 1.048 | 0.137 |

**donutExcl_005 anomaly.** This variant yields negative unfolded signal at $p_T \geq 28\,\text{GeV}$, implying the ABCD background estimate exceeds the tight+iso count in those bins. Looking at the raw `h_unfold_sub_result` across $p_T$: values at $p_T = 22$–$26$ are anomalously large (2–6× nominal) before the bin-11 sign flip to −5.66 and bin-12 −0.81. Truth purity for the same variant is healthy (0.65–0.86), so the pathology is in yield extraction, not purity. Likely an interaction between the tight-iso count and the R-factor estimator at low statistics with the tightest inner $R$ + ownership-excluded outer cone. Recommend: re-run with higher MC statistics (more condor jobs

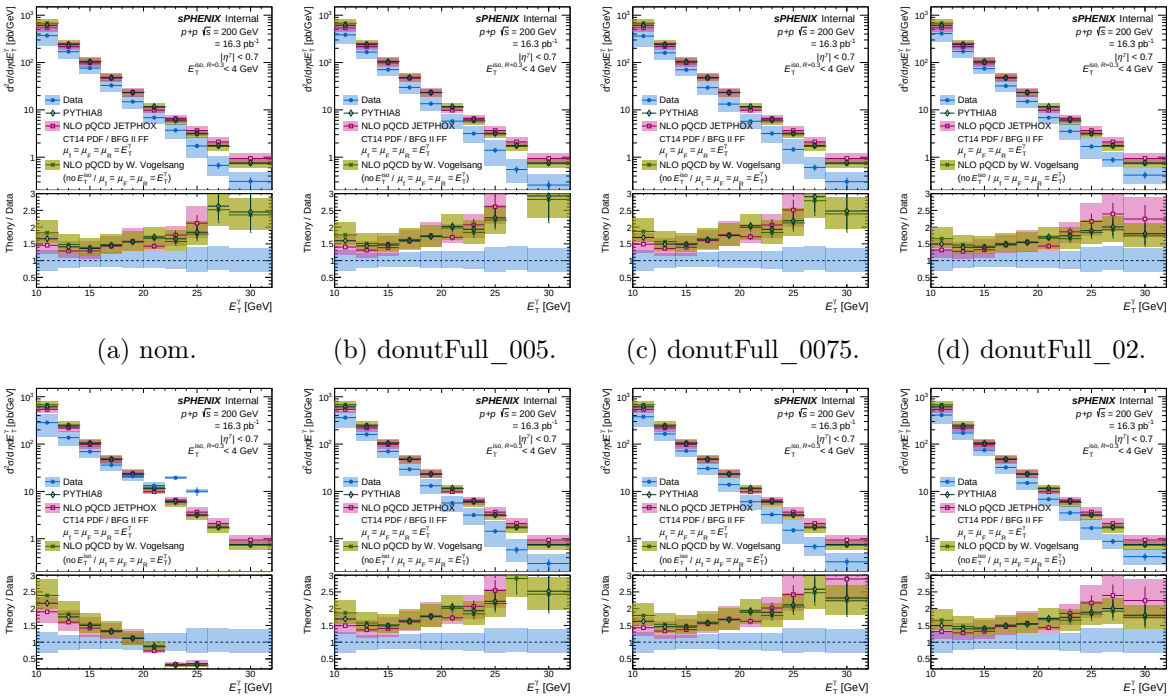
per sample) or drop this variant from the scan.




(a) donutFull / nominal. (b) donutExcl / nominal.

Figure 5: Unfolded-yield ratio (donut / nominal) vs. reco $p_T$. Dashed line at unity. Excluding `donutExcl_005`, variants agree with nominal to $\sim$ 10–14% on $\langle|1-r|\rangle$. The `R_inner`=0.20 variants diverge upward at low $p_T$ (ratio up to 1.35) and stay within $\pm 5\%$ of nominal at mid-$p_T$. `donutExcl_005` is plotted but pT $\geq$ 26 GeV points fall off-scale (negative / $\gg$ 1.4).

## Per-variant final cross section




(a) nom. (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.

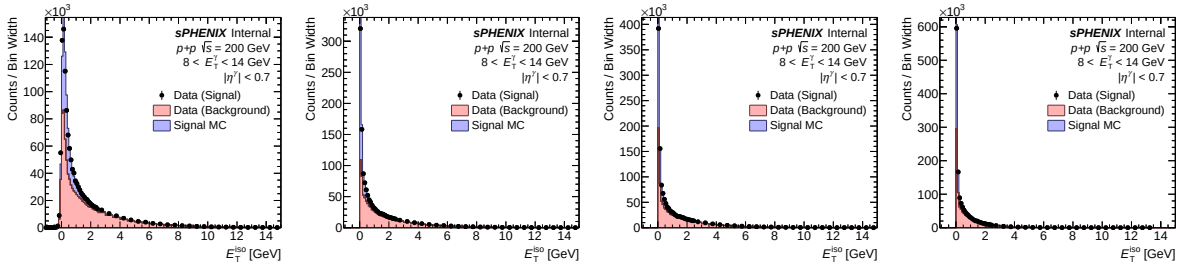
(e) donutExcl_005 (anomalous). (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 6: Per-variant final isolated-photon cross section $d^2\sigma/d\eta\, dp_T$ vs. $p_T^\gamma$, produced by `plot_final_selection.C`. Nominal topo04 iso (first panel) for reference. The `donutExcl_005` panel shows the high-$p_T$ anomaly noted above.

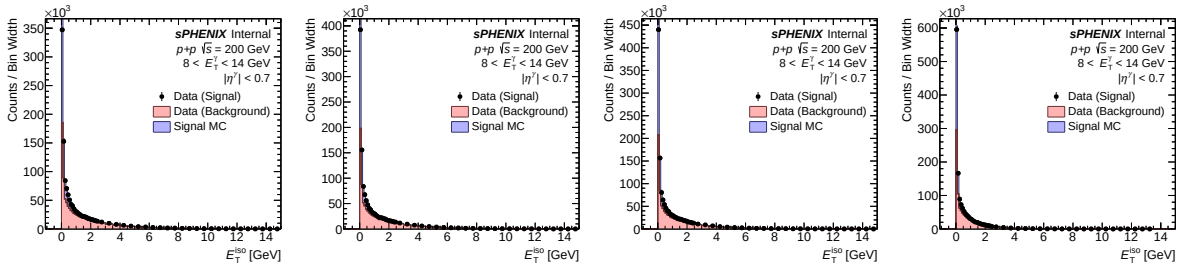
# Isolation ET template fits (CONF)

Produced by `plotting/CONF_plots.C` from `data_histo_bdt_<var>.root` (tight and non-tight data) and `MC_efficiency_bdt_<var>.root` (MC signal template). Two $p_T$ bin ranges shown as representatives: **8–14 GeV** (bins 0–2) and **20–26 GeV** (bins 6–8). The fit overlays the MC signal shape + a non-tight-derived background template onto the tight-region data, extracting the signal fraction per pT range. Full 4-range sets (0–2, 3–5, 6–8, 9–9) for all 7 donut variants are in `plotting/figures/h1D_iso_fit_bdt_donut*.pdf`; the selection report includes them in each per-config section via the standard PLOT_GROUPS "Isolation fit (CONF)" entry.

## Low $p_T$ bin range: $8 < p_T < 14$ GeV (bins 0–2)



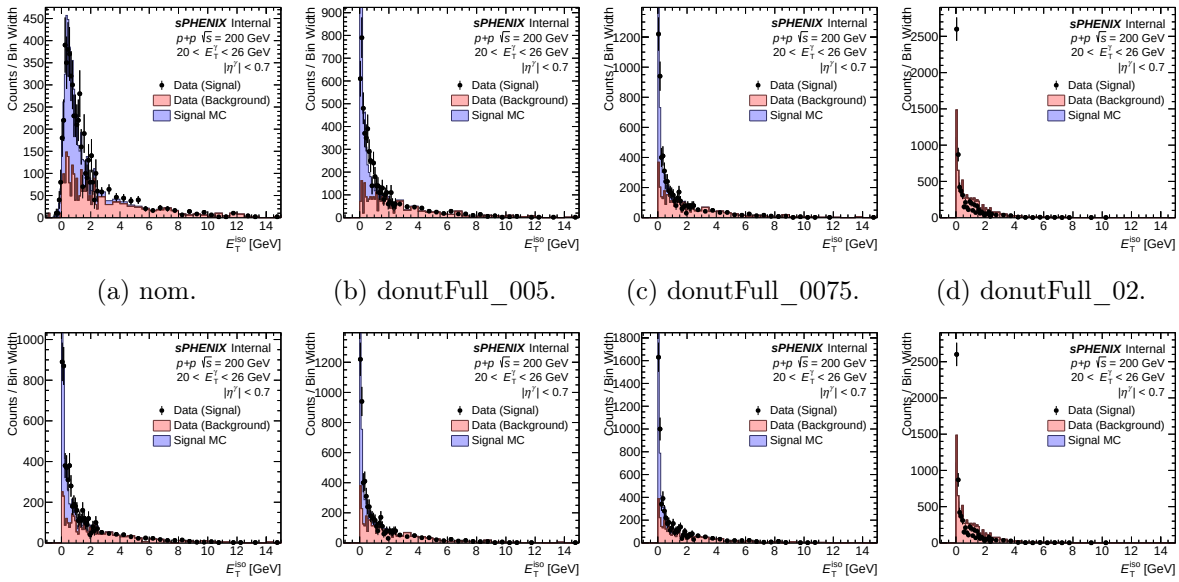

(a) nom. (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.




(e) donutExcl_005. (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 7: Isolation-ET templates vs. tight data (points), with MC signal template (blue fill) and non-tight data-derived background template (red fill), for $8 < p_T < 14$ GeV. Nominal reference and all 7 donut variants.

### High $p_T$ bin range: $20 < p_T < 26$ GeV (bins 6–8)




(a) nom. (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.

(e) donutExcl_005. (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 8: Same as above for $20 < p_T < 26$ GeV. The `donutExcl_005` high-$p_T$ pathology noted in the cross-section section shows up here as the signal+background templates failing to cover the tight-region data shape.

## BDT model alternatives: `base_v0` and `base_v0E`

Cross-check on BDT model choice: two additional variants apply earlier model versions (`base_v0`, `base_v0E`) across both ET bins via `bdt_et_bin_models=["base_v0","base_v0"]` (resp. v0E). Nominal topo04 iso and nominal tight/non-tight BDT cuts are kept. Both $cluster_bdt_CLUSTERINFO_CEM$ and $*_base_v0E$ branches already exist in the slimtree; no re-apply needed. HTCondor cluster 2734512 (2 jobs, ∼20 min each).

### Summary numbers

Table 5: Truth purity, ABCD/truth closure, and unfolded-yield ratio to nominal, at selected $p_T$ bins. Lower $\langle|1-r|\rangle$ is better closure. Higher $|1-r_{\mathrm{xsec}}|$ means larger deviation of the cross section from nominal.

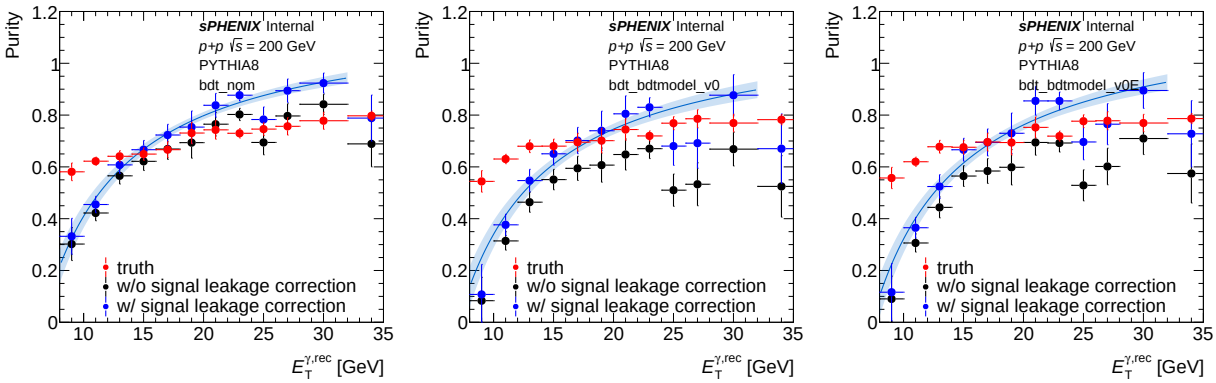
| | Truth purity | | | ABCD/truth closure | | |
|---|---|---|---|---|---|---|
| Variant | $y(9)$ | $y(21)$ | $y(34)$ | $r(9)$ | $r(21)$ | $\langle\lvert 1-r\rvert\rangle$ |
| nom (reference) | 0.581 | 0.743 | 0.797 | 1.795 | 0.910 | 0.185 |
| bdtmodel_v0 | 0.544 | 0.744 | 0.782 | 2.177 | 0.973 | 0.226 |
| bdtmodel_v0E | 0.557 | 0.753 | 0.786 | 2.491 | 0.961 | 0.265 |

Table 6: Unfolded-yield ratio (variant / nominal) across the $p_T$ spectrum.

| Variant | $r(9)$ | $r(15)$ | $r(21)$ | $r(28)$ | $r(34)$ | $\langle\lvert 1-r\rvert\rangle$ |
|---|---|---|---|---|---|---|
| bdtmodel_v0 | 0.751 | 0.980 | 1.014 | 1.183 | 1.604 | 0.196 |
| bdtmodel_v0E | 0.760 | 0.974 | 1.006 | 1.219 | 1.742 | 0.223 |

**Observations.** (1) Truth purity for the v0/v0E models is close to (but slightly lower than) nominal at low $p_T$ (0.54–0.56 vs. 0.58), and essentially identical at high $p_T$. (2) **ABCD closure is worse** for both alternatives: $\langle|1-r|\rangle = 0.226$ (v0) and 0.265 (v0E) vs. 0.185 nominal. The low-$p_T$ over-prediction is particularly strong ($r = 2.18$ and 2.49 at $p_T = 9\,\text{GeV}$ vs. 1.80 nominal), suggesting the ABCD R-factor assumption is less satisfied with these BDT discriminants. (3) Cross section at high $p_T$ diverges sharply from nominal: 1.60× and 1.74× at $p_T = 34\,\text{GeV}$ respectively. The v0/v0E models either admit substantially more tight+iso yield (looser BDT) or have larger non-closure absorbing into the subtracted "signal" yield. Given that the cross-section ratio grows with $p_T$ while closure degrades at low $p_T$, the high-$p_T$ excess is likely driven by poorer signal–background separation of the older BDT variants in the region where statistics are thin.

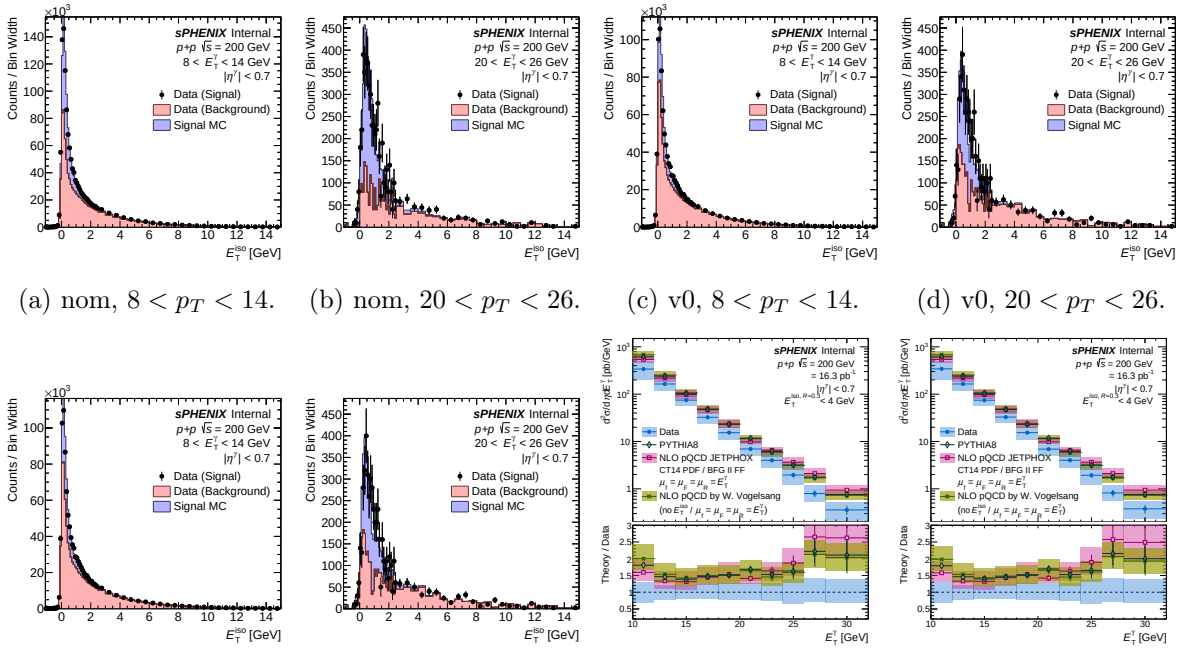
## Per-variant MC purity closure




(a) nom (reference). (b) bdtmodel_v0. (c) bdtmodel_v0E.

Figure 9: MC purity closure for the two alternative BDT models. Each panel shows the ABCD-derived purity (points), the fit, and the truth-matched purity (red markers).

## Isolation ET template fits




(a) nom, $8 < p_T < 14$. (b) nom, $20 < p_T < 26$. (c) v0, $8 < p_T < 14$. (d) v0, $20 < p_T < 26$.

(e) v0E, $8 < p_T < 14$. (f) v0E, $20 < p_T < 26$. (g) Final $d^2\sigma$, v0. (h) Final $d^2\sigma$, v0E.

Figure 10: CONF template overlays (`h1D_iso_*`: tight data points, MC signal template in blue fill, non-tight background template in red fill) for nominal and the two alternative BDT models at two $p_T$ ranges, followed by the unfolded final cross section per alternative model. Poorer template agreement for v0/v0E at high $p_T$ is consistent with the weaker closure reported above.